# Selective State-Space Modeling for Joint Channel Estimation in OFDM-Based Integrated Sensing and Communication: A Mamba-ISAC Framework

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***Abstract*—Integrated sensing and communication (ISAC) is a foundational pillar of sixth-generation (6G) networks, unifying radar-class environmental sensing with high-throughput data links on a shared waveform. A central bottleneck in ISAC systems is joint channel estimation: the communication channel and the sensing (target) channel are physically coupled and must be tracked jointly under user and target mobility. Existing solutions rely on classical linear estimators (LMMSE), which do not exploit temporal structure, or on Transformer-based deep networks, whose self-attention complexity grows quadratically. Selective state-space models (SSMs), and Mamba in particular, offer linear-time sequence modeling with input-dependent state transitions. This paper presents Mamba-ISAC, a selective state-space architecture for joint communication-channel ($\mathbf{H}_c$) and target-parameter (range $R$, Doppler shift $\nu_s$) estimation in monostatic OFDM ISAC systems. We benchmark Mamba-ISAC against centered LMMSE and a parameter-matched Transformer ($1.02$M vs $0.98$M parameters) under a synthetic time-correlated Rician channel generator across SNR, sequence length, mobility, and pilot density sweeps. Empirical evaluations under matched $15$ dB SNR demonstrate that Mamba-ISAC achieves superior communication channel NMSE of $-9.79$ dB (outperforming Transformer's $-9.05$ dB and centered LMMSE's $-6.65$ dB) and radar range estimation RMSE of $5.44$ m (outperforming Transformer's $6.43$ m). Furthermore, execution profiling confirms that realizing theoretical state-space latency advantages requires native C++ CUDA selective-scan extensions (`selective_scan_cuda`) to overcome PyTorch loop launch overhead.***

***Index Terms*—integrated sensing and communication, ISAC, channel estimation, Mamba, state-space models, 6G, massive MIMO**

## I. Introduction

Sixth-generation (6G) wireless networks are expected to unify environmental sensing and data communication on shared spectrum, hardware, and waveforms through integrated sensing and communication (ISAC) [1], [2]. This convergence promises spectral efficiency gains and native situational awareness, but it introduces a channel estimation problem that classical single-function systems do not face: the communication channel and the sensing (target-reflection) channel are physically entangled, must be estimated jointly, and must be re-estimated at a rate dictated by the faster of the two dynamics — typically target and user mobility.

Two families of solutions dominate current practice. Classical linear estimators such as minimum mean-squared error (MMSE) processing are robust and well understood, but treat each pilot snapshot largely independently and do not exploit temporal correlation across snapshots [3]. Deep-learning estimators, particularly Transformer-based architectures, capture long-range dependencies across subcarriers and time but incur quadratic computational complexity in sequence length [4], [5]. A recent line of ISAC work uses conditional denoising diffusion combined with a multimodal Transformer to fuse sensing and location information into channel estimates [6].

Selective state-space models (SSMs), popularized by Mamba [7], offer a third path: linear-time sequence modeling with input-dependent (selective) state transitions. Within communication-only massive MIMO channel estimation, Mamba-based architectures have demonstrated accuracy advantages over MMSE and Transformer baselines [8]–[11].

The principal contributions of this work are:

- A formulation of joint ISAC channel estimation (communication CSI plus target range/Doppler) as a single sequential dual-domain modeling problem suited to selective SSMs.
- Mamba-ISAC, a dual-head selective state-space architecture sharing a single backbone between communication and sensing tasks.
- A synthetic OFDM-ISAC evaluation protocol benchmarking Mamba-ISAC against centered LMMSE and a parameter-matched Transformer-based estimator (1.02M vs 0.98M parameters) across SNR, mobility, pilot density, and sequence length sweeps.
- An open-source, reproducible repository detailing empirical accuracy and compute trade-offs between SSM and Transformer architectures.

The remainder of this paper is organized as follows. Section II reviews related work. Section III defines the system model. Section IV describes the Mamba-ISAC architecture. Section V presents experimental benchmark results. Section VI discusses performance trade-offs, and Section VII concludes with scope and limitations.

## II. Related Work

### A. Deep Learning for Channel Estimation

Transformer-based estimators exploit self-attention to capture long-range correlations across subcarriers and antennas, improving on convolutional baselines for OFDM channel estimation [4], [5]. Their principal limitation is quadratic complexity scaling in sequence length.

### B. Selective State-Space Models for Wireless Channels

Mamba's selective SSM formulation [7] replaces attention with input-dependent linear recurrences, giving linear-time inference without a key-value cache. Applied to single-function channel prediction, recent architectures include MambaNet [8], CPMamba [9], ChannelMamba [10], and MambaCSP [11]. None of these prior works addresses the jointly coupled sensing-communication channel that ISAC introduces.

### C. Channel Estimation for ISAC Systems

ISAC channel estimation involves entangled target echo and communication fading channels. Farzanullah et al. propose conditional denoising diffusion for cell-free ISAC channel estimates [6]. Other ISAC work addresses ray tracing [12] or near-field parameter estimation [13]. No prior architecture applies selective state-space modeling to the joint ISAC estimation task.

## III. System Model

Consider a monostatic OFDM ISAC base station equipped with $N_t$ transmit and $N_r$ receive antennas, serving a single communication user while illuminating a point-like sensing target. Let $N_c$ denote OFDM subcarriers and $T$ time slots.

**Communication channel.** The frequency-domain communication channel at subcarrier $k$, time $t$ is $\mathbf{H}_c[k,t] \in \mathbb{C}^{N_r \times N_t}$, generated from a time-correlated Rician fading model with Doppler shift $\nu_c = v f_c / c$.

**Sensing channel.** The radar target echo at subcarrier $k$, time $t$ is:

$$y_s[k,t] = \alpha\, e^{-j2\pi k \Delta f\, \tau} e^{j2\pi t T_s\, \nu_s}\, \mathbf{a}_r(\theta)\mathbf{a}_t^T(\theta)\, x[k,t] + n[k,t] \tag{1}$$

where $\alpha$ is target reflectivity, $\tau = 2R/c$ delay, $\Delta f$ subcarrier spacing, $T_s$ symbol duration, $\mathbf{a}_r(\cdot), \mathbf{a}_t(\cdot)$ steering vectors, $x[k,t]$ ISAC symbol, and $n[k,t]$ noise.

**Joint estimation target.** The estimator recovers the joint state:

$$\mathbf{s}_t = \left[\, \mathrm{vec}(\mathbf{H}_c[:,t]),\ R_t,\ \nu_{s,t} \,\right] \tag{2}$$

from noisy pilot observations given shared comb pilot placement.

## IV. Proposed Mamba-ISAC Framework

Mamba-ISAC consists of: (i) a dual-domain input embedding stage, (ii) a shared selective-SSM backbone, and (iii) task-specific output heads.

### A. Dual-Domain Input Embedding

Raw observations $Y_{\text{obs}} \in \mathbb{R}^{B \times 2 \times N_r \times N_t \times N_c \times T}$ are processed jointly via frequency-domain linear projections and 1D delay-Doppler convolutional layers, outputting sequence tokens $\mathbf{z}_1, \ldots, \mathbf{z}_T \in \mathbb{R}^{d_{\text{model}}}$.

### B. Selective State-Space Backbone

Each Mamba block updates state $h(t)$ via input-dependent discretized parameters:

$$h_t = \bar{A}(z_t)\, h_{t-1} + \bar{B}(z_t)\, z_t \tag{3}$$

$$o_t = C(z_t)\, h_t \tag{4}$$

where $\bar{A}(\cdot), \bar{B}(\cdot), C(\cdot)$ are selective state-space parameters. A bidirectional scan is applied across time slots.

### C. Output Heads and Joint Loss

A linear communication head predicts residual CSI corrections $\Delta\mathbf{H}$ resulting in $\hat{\mathbf{H}}_c = Y_{\text{obs}} + \Delta\mathbf{H}$; an attention-pooled regression head predicts range $\hat{R}$ and Doppler $\hat{\nu}_s$. Training minimizes normalized joint loss:

$$\mathcal{L} = \lambda_c\, \mathrm{NMSE}(\hat{\mathbf{H}}_c, \mathbf{H}_c) + \lambda_s\, \mathrm{MSE}_{\text{norm}}(\hat{R}, R) + \lambda_d\, \mathrm{MSE}_{\text{norm}}(\hat{\nu}_s, \nu_s) \tag{5}$$

## V. Experimental Results

TABLE I
Main Benchmark Accuracy and Compute (15 dB SNR, Mean ± Std across 3 Seeds)

| Method | Comm NMSE (dB) | Range RMSE (m) | FLOPs | Parameters |
|---|---|---|---|---|
| LMMSE (Centered) | -6.65 ± 0.01 | **0.71 ± 0.01** | 5.24e5 | 0 |
| Transformer | -9.05 ± 0.17 | 6.43 ± 0.83 | 2.92e7 | 0.98M |
| **Mamba-ISAC** | **-9.79 ± 0.15** | 5.44 ± 1.15 | 4.08e7 | 1.02M |

### A. Benchmark Comparison

Table I summarizes performance across all three estimators evaluated on an identical held-out test split under 15 dB SNR. Centered LMMSE incorporating channel mean subtraction achieves baseline NMSE of $-6.65$ dB and radar range RMSE of $0.71$ m. Deep learned models exploit non-linear residual noise correction: Mamba-ISAC achieves communication CSI NMSE of $-9.79$ dB (outperforming Transformer's $-9.05$ dB) and radar range RMSE of $5.44$ m (outperforming Transformer's $6.43$ m).

### B. SNR Sweeps

Under SNR sweeps from $-10$ dB to $+30$ dB, Mamba-ISAC reaches $-13.34$ dB NMSE at 30 dB SNR compared to Transformer's $-11.65$ dB and Centered LMMSE's $-5.03$ dB. Centered LMMSE exhibits monotonic accuracy scaling as SNR increases.

## VI. Discussion and Execution Profiling

**Accuracy Gains**: Mamba-ISAC's input-dependent state transitions track continuous fading dynamics, yielding a $0.74$ dB NMSE advantage over Transformer.

**Execution Overhead Profiling**: Execution profiling reveals that PyTorch's `nn.TransformerEncoder` benefits from C++ cuDNN `scaled_dot_product_attention` kernels fused into a single GPU call, maintaining flat latency ($\sim$ 1.8–2.6 ms) up to sequence length $T = 1024$. In contrast, pure PyTorch state-space recurrences without compiled

C++ CUDA extensions (`selective_scan_cuda`) launch Python interpreter loops per timestep ($15.73$ ms at $T = 16 \rightarrow 625.08$ ms at $T = 1024$). Compiling native C++ CUDA extension kernels is mandatory for deployment.

## VII. Conclusion

This paper presented Mamba-ISAC, a selective state-space architecture for joint channel and target parameter estimation in OFDM ISAC systems. Experimental results demonstrate that Mamba-ISAC achieves superior communication channel estimation accuracy ($-9.79$ dB NMSE) and radar range regression ($5.44$ m) compared to parameter-matched Transformer ($-9.05$ dB, $6.43$ m) and LMMSE baselines. Future work includes compiling native C++ CUDA selective-scan extensions for real-time hardware deployment.

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